VISIBLE ONE LIMITED

Scenario Assessment

IT Manager (Infrastructure, DevOps & Security) — Candidate Brief

> **What this is.** A single realistic scenario, submitted as a written package. It is designed to show us how you think about cloud architecture, automation and risk — and how clearly you explain your thinking to people who are not engineers. It is deliberately under-specified in places. Asking good questions, and stating your assumptions clearly, counts in your favour.
>
> **This is the first stage of the process.** Every submission is reviewed before any interview is scheduled. Candidates whose submissions meet the bar are invited to a follow-up interview; we will let you know either way.

1. Format and time

Stage	Time	Detail
Stage 1 — Submission	≤ 4 hours	Your own time, at home. Return your written package within **5 working days** of receiving this brief.
Review	—	Every submission is read and assessed by the panel. We aim to come back to you within one week of the deadline.
Stage 2 — Interview	≈ 45 minutes	By video call, for shortlisted candidates only. You will walk us through your submission and answer questions on it.
Panel	—	Managing Director and the current IT Manager. One is non-technical — write and speak accordingly.

> **Please do not spend more than four hours.** We mean this. We are assessing judgement, not endurance, and we will not score a 20-hour submission higher than a focused 4-hour one. If you run short of time, prioritise the architecture and the reasoning, and tell us in one line what you would have done with more time — that itself is a useful signal.

Use of AI tools

You may use AI assistants for this exercise. We use them ourselves. The requirement is simply that the reasoning is yours: submit a design you understand and stand behind, rather than the most elaborate one you can assemble. A simpler architecture you have thought through is worth more to us than a sophisticated one you have not.

2. The scenario

Client background

Northpoint Living is a Hong Kong homeware and lifestyle retailer. They sell online and through two physical stores. They have been our client for four years, and have just told us they intend to expand into Singapore and Taiwan within 18 months.

Their online store currently runs on a single virtual server managed by a local hosting provider. It works, but it has failed twice in the last twelve months — both times during a major sales campaign, which is the worst possible moment and the reason they are now willing to spend money on the problem.

Current state

Item	Detail
Platform	Open-source PHP e-commerce application on a CMS, single instance
Server	One VPS — 8 vCPU, 16 GB RAM. Application, database and file storage all on the same host
Catalogue	~8,200 active SKUs, ~180 GB of product media
Traffic	~45,000 sessions/month steady; ~5× through November–December
Campaign peaks	Two per year, roughly **10× normal traffic for a 4–6 hour window**
Payments	Redirect to a third-party payment gateway. No card data is stored by the application
Customer data	Name, email, phone, delivery address, order history — held in the application database
Deployment	Manual file upload by the developer. No staging environment, no version control on the server
Backups	Nightly archive to the same hosting provider. Never test-restored
Monitoring	None, beyond the client telephoning us
Measured uptime	~99.2% over the last 12 months

What the client wants

Requirement	Target
Availability	**99.9%** monthly, measured on the storefront and checkout
Recovery point (RPO)	≤ **15 minutes** of order data
Recovery time (RTO)	≤ **60 minutes** to restore service
Peak handling	Absorb a 10× spike **without manual intervention on the night**
Deployments	Zero-downtime, repeatable, with a rollback path
Data residency	Customer personal data to stay in Hong Kong, or the closest available region. Justify your choice
Compliance	Hong Kong PDPO applies to customer personal data
Budget	Target **HKD 18,000 per month** steady state. Campaign-period overspend is acceptable if you justify it
Roadmap	Singapore and Taiwan storefronts within 18 months

Choice of cloud

You choose the platform — AWS, Azure, Alibaba Cloud, Google Cloud, or a hybrid. There is no right answer and we are not testing loyalty to a vendor. We are testing whether you can justify the choice against *this* client's constraints: the region requirement, the budget, the expansion plan, and what a small team can realistically operate.

3. What to produce

Six items. Keep the whole package to roughly **eight pages**. Concise and clear beats comprehensive.

A. Architecture diagram

1 PAGE · ANY TOOL, INCLUDING HAND-DRAWN AND PHOTOGRAPHED

The target-state design. Show network boundaries, availability zones or equivalent, where data lives, and how traffic reaches the application. Legibility matters more than beauty.

B. Design rationale

MAX 3 PAGES

Why this design. We are specifically interested in:

- How you meet the 99.9% target, and where the design can still fail.
- How the database survives losing a node without breaching the 15-minute RPO.
- How the 10× campaign spike is absorbed, and how quickly the scaling reacts.
- **What you deliberately left out of phase one, and why.**

C. Cost estimate

1 PAGE · TABLE FORMAT

Monthly steady-state cost and estimated campaign-month cost, itemised by component, in HKD or USD. Approximate figures from public pricing are fine — state your assumptions. If you exceed the target budget, say so plainly and explain what the extra money buys.

D. Automation

1–2 PAGES

Two parts:

- A **representative excerpt** of infrastructure-as-code or pipeline configuration — one meaningful component, not the whole stack, and it does not need to run. We want to see how you write it.
- A short description of what you would automate **first** after go-live, and what you would deliberately leave manual.

E. Migration and rollback plan

1 PAGE

How you move from the current single server to the new platform with minimal disruption. Include the cutover sequence, how you handle orders placed during the switch, your rollback trigger, and who decides to pull the trigger.

> **F. Risk register**
>
> HALF A PAGE · TABLE FORMAT
>
> Your top five risks in this project, with likelihood, impact and mitigation. At least one should be a risk to *us* as the agency, not only to the client.

4. After you submit

Your package is read and assessed by the panel before any interview is arranged. We will come back to you either way, and we aim to do so within one week of the submission deadline.

If you are shortlisted, we will invite you to a **video call of roughly 45 minutes** in which you will walk us through your submission and answer our questions on it. You will share your screen. Slides are optional — a clear diagram and a verbal walkthrough do the job perfectly well. Please join from somewhere quiet with a stable connection; if we lose you mid-conversation, reconnect and we will pick up where we stopped. A dropped connection is a technical problem, not a mark against you.

5. How your submission is assessed

The weightings are shown so that you can spend your four hours in the right places.

Dimension	Weight	Where it shows
Cloud architecture and high availability	**25%**	Deliverables A, B
Automation and infrastructure-as-code	**20%**	Deliverable D
Communication and clarity	**15%**	Throughout the written package
Cost and commercial judgement	**15%**	Deliverable C
Security and data protection	**15%**	Throughout, plus F
Operability — monitoring, backup, recovery	**10%**	Deliverables B, E

What we are not looking for

- **A working deployment.** Do not build anything. Do not spend money.
- **Certainty you do not have.** "I would need to test this" is a legitimate and well-received answer.
- **Vendor marketing language.** Tell us what the component does for this client, not what the brochure says.

6. Practical notes

- Send your package as a **single PDF** to the address given in your email from us, by the deadline stated there.
- Questions about the brief are welcome and are not held against you — email us. If we do not answer in time, state your assumption in the document and proceed.
- Northpoint Living is a fictional composite. Do not research it.
- **Do not include anything confidential from a current or former employer** — no client names, diagrams, code or pricing. We will treat that as a serious negative signal, for reasons we hope are obvious.

- Your materials are used solely for this assessment, are seen only by the interview panel, and are deleted within 90 days of the process concluding.
